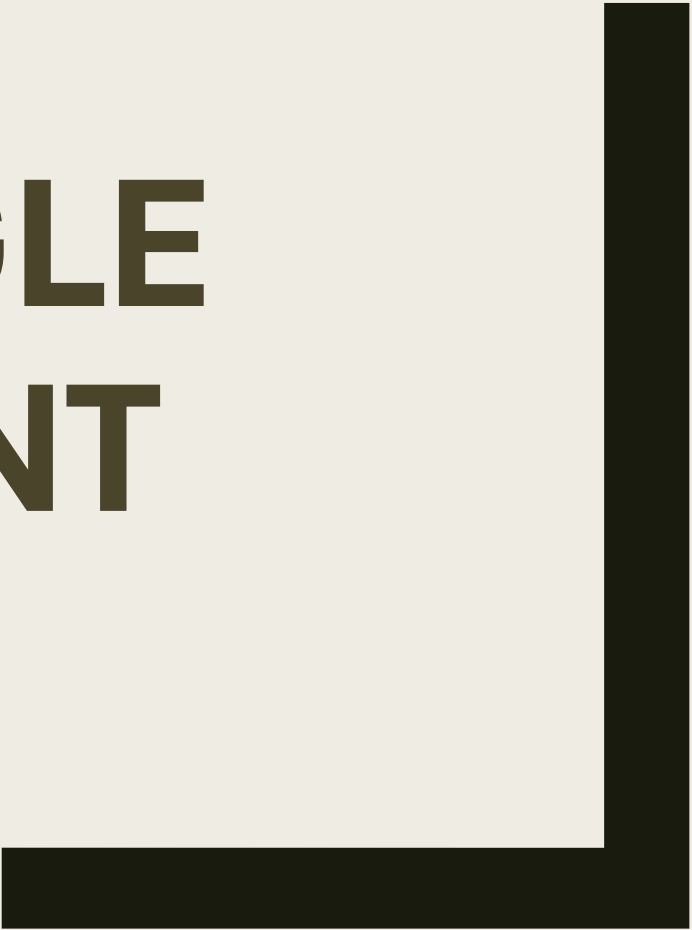




STATIC SINGLE ASSIGNMENT



Static Single Assignment (SSA) IR

Consider the following:

$$x = y + z$$

$$x = x + 1$$

$$w = y + z$$

$$v = x + 3$$

Static Single Assignment (SSA) IR

Consider the following:

$$x_1 = y + z$$

$$x_2 = x_1 + 1$$

$$w_1 = y + z$$

$$v_1 = x_2 + 3$$

Why SSA?

1. Optimizations algorithms become simpler if each variable has only one definition
2. Unrelated uses of same variable become independent
3. More values become available at each program point

Converting to SSA

1. Replace the target of each assignment with a new variable
2. Replace each use of a variable with the **version** of the variable **reaching** that point

Converting to SSA

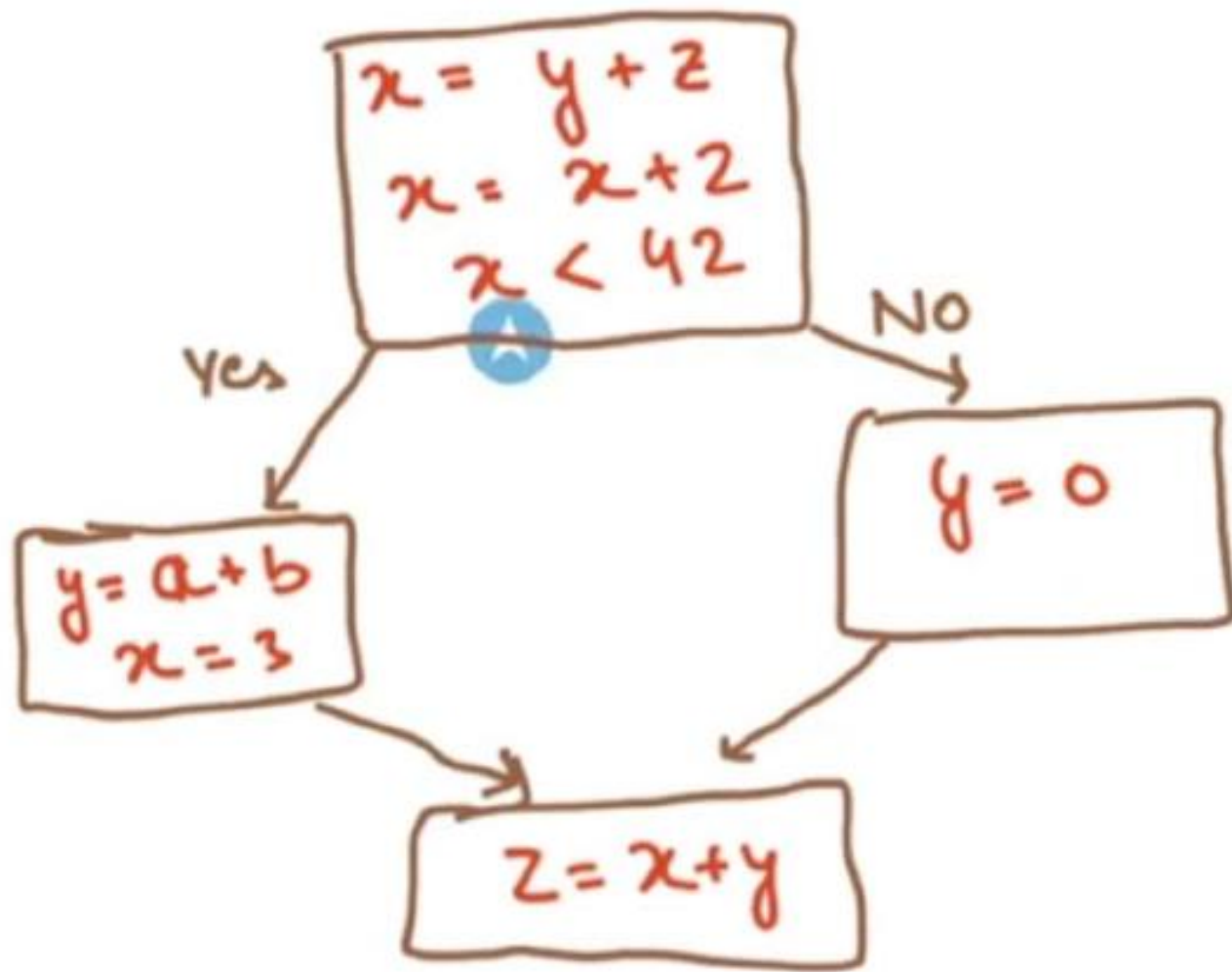
$$x = y + z$$

$$a = b + x$$

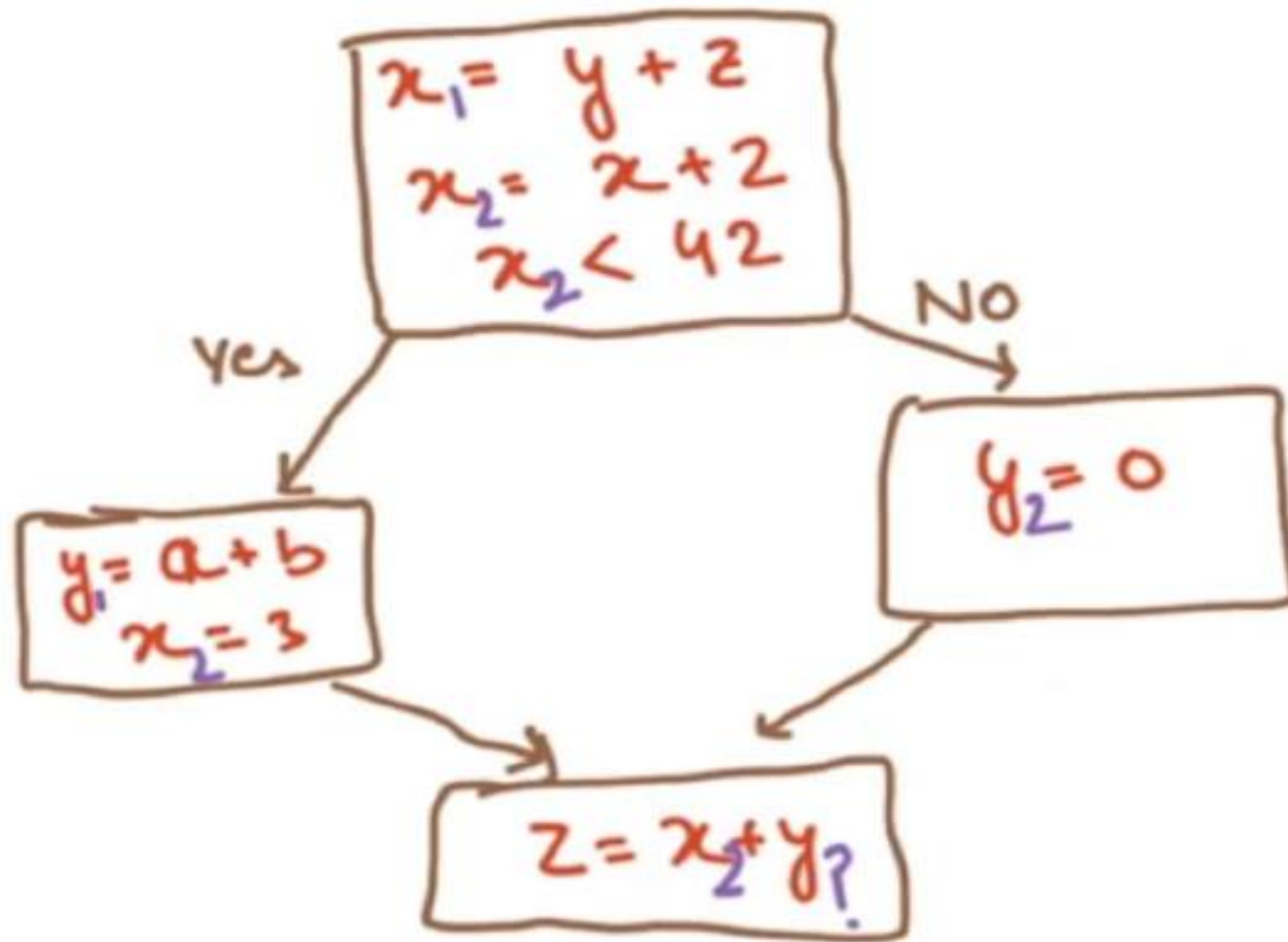
$$x = a + 3$$

$$y = x - a$$

Converting to SSA



Converting to SSA



Basic Block

A basic block is a maximal sequence of instructions with

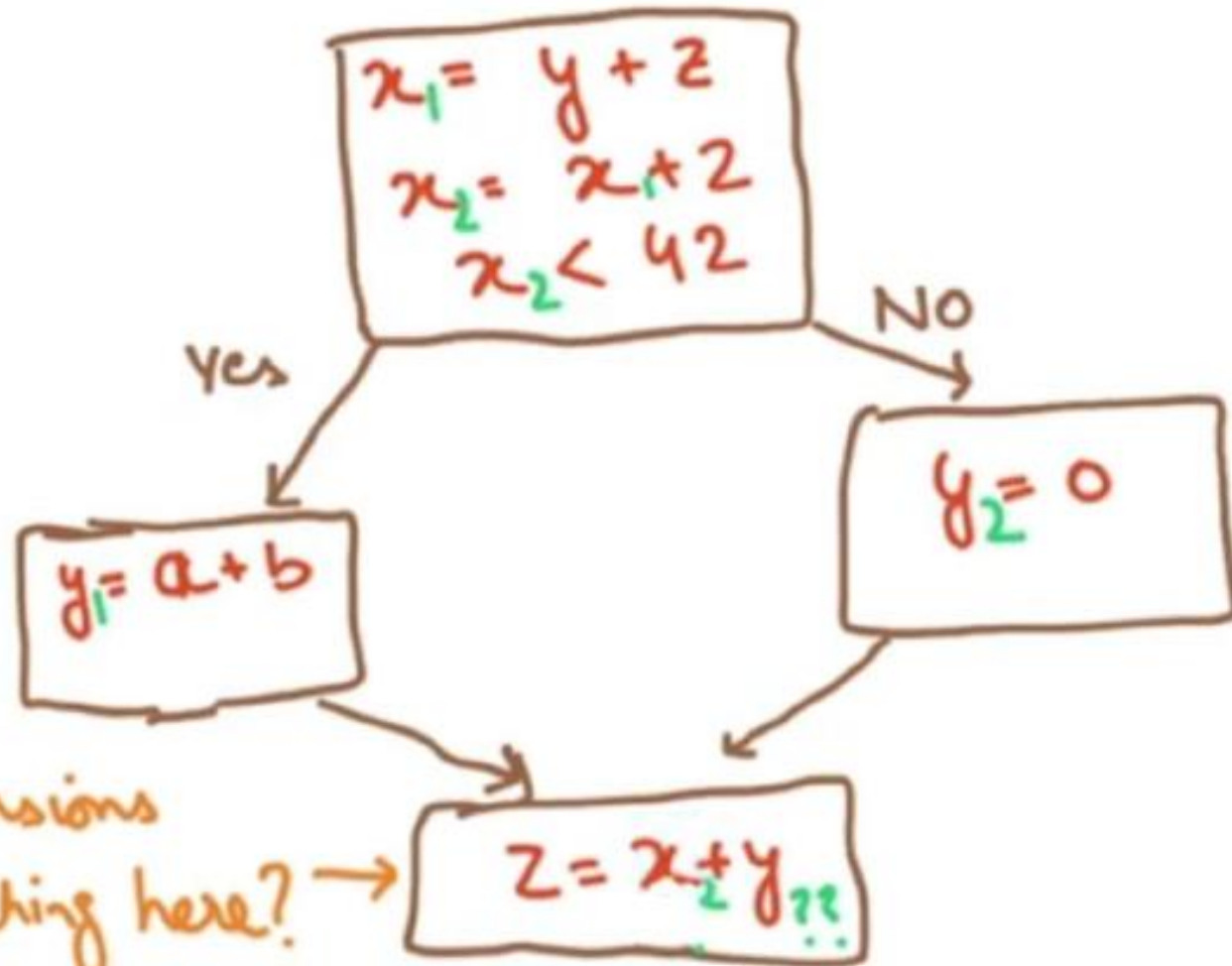
- no labels (except at first instruction)
- no jumps (except in the last instruction)

Basic Block

Idea:

- Cannot jump into a basic block
(except at beginning)
- Cannot jump out of a basic block
(except at end)
- Single-entry single-exit
straight-line code segment

Converting to SSA



What versions
are reaching here? →

Converting to SSA

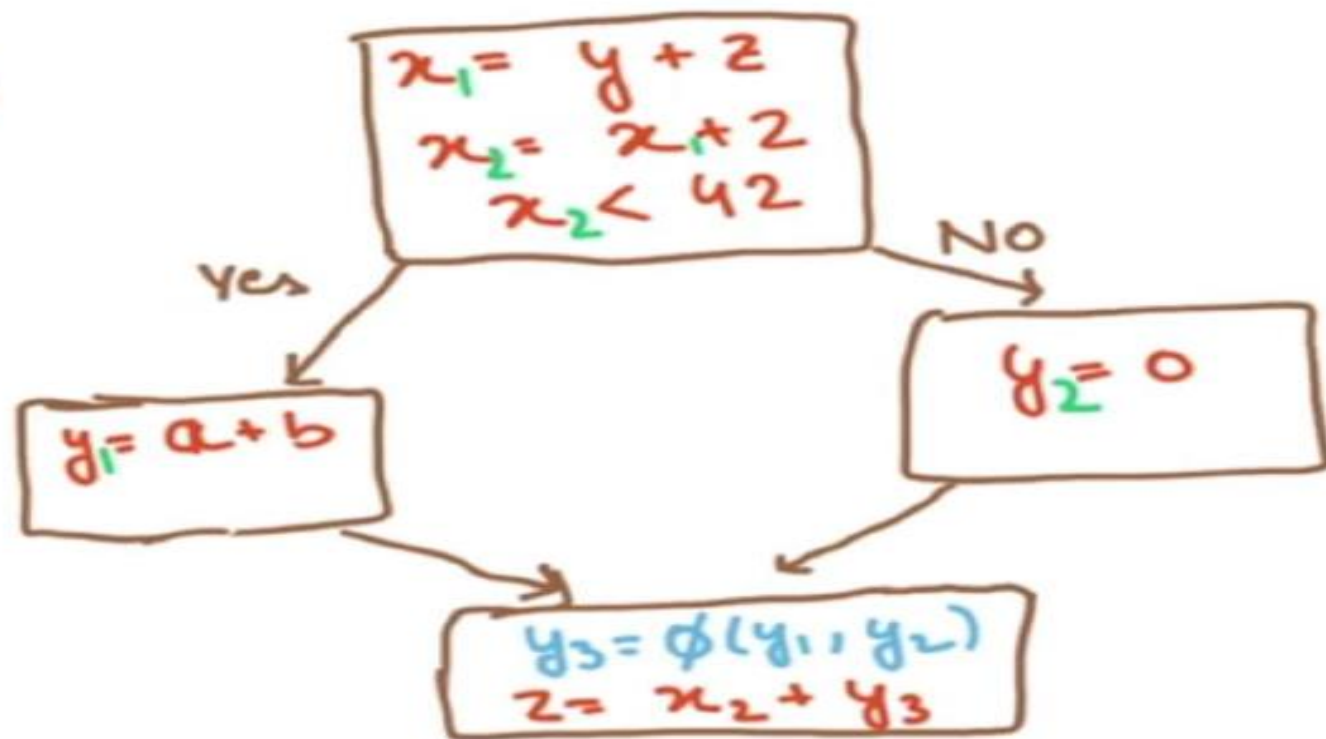
Solution:

Φ (PHI)

node

aka

Φ
function



PHI nodes

1. Φ function chooses the version depending on the incoming edge
2. Present only at the beginning of a basic block